Lecture 6: Unsupervised learning

Vegard Høghaug Larsen

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Outline

- What is Unsupervised Learning?
- Dimensionality Reduction
- Principal Component Analysis (PCA)
- K-Means Clustering

What is Unsupervised Learning?

Definition: Unsupervised Learning is a machine learning paradigm in which the model learns patterns from data that does not have labeled responses. Instead of predicting an outcome, the focus is on discovering inherent structures within the data.

- No pre-assigned labels are needed.
- The goal is to uncover hidden patterns, clusters, or associations.
- Widely used for exploratory data analysis.

Key Concepts in Unsupervised Learning

- **Dimensionality Reduction:** Reducing the number of variables while preserving essential information (e.g., PCA, t-SNE).
- **Clustering:** Grouping data points based on similarity (e.g., K-Means, Hierarchical clustering).
- **Density Estimation:** Modeling the distribution of data to identify areas of high or low concentration.
- **Anomaly Detection:** Identifying rare items, events, or observations that raise suspicions by differing significantly from the majority.

Comparison with Supervised Learning

Unsupervised Learning

- Uses unlabeled data.
- Focuses on pattern discovery.
- Commonly applied in clustering, dimensionality reduction, and anomaly detection.

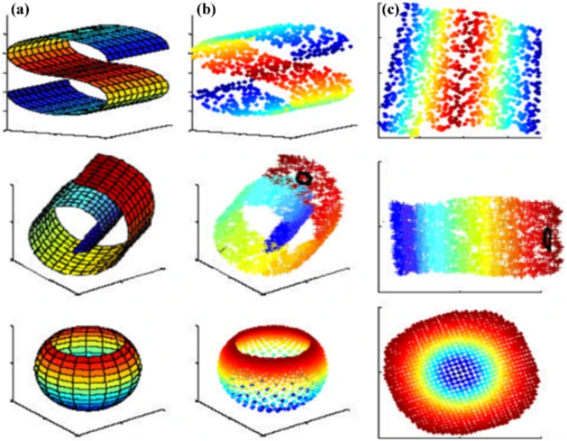
Supervised Learning

- Uses labeled data.
- Focuses on prediction and classification tasks.
- Examples include regression, classification, and forecasting.

Introduction to Dimensionality Reduction

- **What?** Reducing the number of variables/features.
- **Why?** The "Curse of Dimensionality" (overfitting, computation, visualization).
- **Benefits?** Simpler models, faster computation, visualization.

Dimensionality reduction



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Principal Component Analysis (PCA)

Principal Component Analysis (PCA): Concept

- Linear dimensionality reduction technique.
- Finds new axes (principal components) that maximize variance.
- Can think of it as a decomposition of the data where we keep the most important information.
- Orthogonal components: Each captures remaining variance.
- Unsupervised: Ignores class labels.

PCA: Applications and Use Cases

- **Data Visualization (2D/3D Projections):**
  - PCA reduces high-dimensional data to 2 or 3 dimensions.
  - Facilitates the visualization of complex data structures.
  - Helps to identify clusters, trends, and outliers visually.
- **Preprocessing: Feature Reduction and Noise Filtering:**
  - Reduces the number of features, making subsequent models simpler and faster.
  - Mitigates the *curse of dimensionality* by focusing on the most informative aspects of the data.
  - Filters out noise by discarding components with low variance.
- **Compression: Image Compression and Beyond:**
  - *Image Compression:*
    - In techniques like *Eigenfaces*, PCA represents facial images using a reduced set of features.
    - Maintains essential image characteristics while significantly reducing file size.
  - Useful for other data types where reducing storage and speeding up processing are crucial.

Limitations of PCA

- **Linearity:** PCA is limited in capturing non-linear data.
- **Feature Scaling:** Standardization is crucial for unbiased results.
- **Interpretability:** Can be difficult with many features
- **Information Loss:** Must use explained variance to determine acceptable amount of loss.
- **Outliers:** Can significantly alter results.

K-Means Clustering

K-Means Clustering: Concept

- **Goal:** Partition data into k clusters such that points within each cluster are as similar as possible, while points in different clusters are as dissimilar as possible.
- **Centroids:**
  - Each cluster is represented by its centroid, which is the mean of the data points assigned to that cluster.
  - The centroid is a point in the feature space that minimizes the squared distance to all points in its cluster.
- **Assignment Rule:**
  - Each data point is assigned to the cluster with the nearest centroid.
  - Distance is typically measured using the Euclidean metric.
- **Hard Clustering:**
  - Each point is assigned exclusively to one cluster (no overlapping clusters or soft membership).

K-Means: Objective

- **Objective:** Minimize the Within-Cluster Sum of Squares (WCSS)
  - Formally, the goal is to partition the data into k clusters $\{S_1, S_2, \ldots, S_k\}$ by minimizing:

$$\min_{S_1,\ldots,S_k} \sum_{i=1}^{k} \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \mu_i\|^2,$$

    where μ_i is the centroid (mean) of cluster S_i.
  - This objective ensures that data points within each cluster are as close as possible to the cluster's centroid, leading to more compact and well-defined clusters.
- **Centroid Definition:**
  - The centroid of a cluster S_i is computed as the mean of all data points in that cluster:

$$\mu_i = \frac{1}{|S_i|} \sum_{\mathbf{x} \in S_i} \mathbf{x}.$$

  - It represents the "center" of the cluster in the feature space and is used as the reference point for assigning points to clusters.

K-Means: Practical Considerations

- **Initialization Sensitivity:**
  - The choice of initial centroids can greatly affect the convergence and final clusters.
  - Use **k-means++** to select well-spaced initial centroids.
  - Run the algorithm multiple times with different random seeds and choose the best outcome based on the objective function.
- **Choosing k:**
  - Use the **Elbow Method** to plot the Within-Cluster Sum of Squares (WCSS) against different values of k and identify a point of diminishing returns.
  - Apply **Silhouette Analysis** to assess the quality of clustering by measuring how similar an object is to its own cluster versus other clusters.
- **Outliers:**
  - Outliers can significantly distort centroid positions and degrade clustering quality.
  - Consider pre-processing steps like outlier detection or robust clustering methods.

K-Means: Applications and Use Cases

- Market segmentation.
- Document clustering.
- Image segmentation/compression.
- Biology and medicine.
- Anomaly detection (points far from centroids).
- General exploratory analysis.

Limitations of K-Means

- **Choosing k:** Requires prior knowledge of structure in the data.
- **Cluster shape and size:** Assumes spherical clusters of the same size.
- **Sensitivity to initialization:** Different starting points can lead to different results.
- **Outliers:** Can significantly alter cluster center points.

Combining PCA and K-Means

- PCA as preprocessing for K-Means.
- Benefits:
  - Reduces dimensionality and noise.
  - Speeds up K-Means.
  - Can improve cluster quality.
  - Facilitates visualization.
- Example: Image clustering.
- Caution: Don't reduce dimensions **too** much.

Conclusion

- **PCA:** Dimensionality reduction, maximizes variance.
- **K-Means:** Clustering, minimizes within-cluster variance.
- **Use PCA for:** High-dimensional data, visualization, noise reduction.
- **Use K-Means for:** Finding groups, exploratory analysis.
- **Powerful combination:** PCA + K-Means.